

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SECURITY INCIDENT MAPPING (SIM) SYSTEM

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the software requirements for the **Security Incident Mapping (SIM)** system. This document provides a detailed description of the system's objectives, functional requirements, non-functional requirements, user interactions, and operational constraints. It serves as a blueprint for the design, development, testing, and evaluation of the application while ensuring that the system meets the needs of its intended users.

1.2 Scope

Security within and around tertiary institutions remains a major concern. Incidents such as theft, physical assault, burglary, sexual assault, and other crimes continue to threaten students, staff, and residents living around campus. Unfortunately, many incidents are either reported late or not reported at all because existing reporting methods are slow, inconvenient, or ineffective.

The Security Incident Mapping (SIM) System is a web-based application developed to provide a simple, secure, and centralized platform through which users can report security incidents quickly. The application enables users to select the type of incident, specify the location, provide a detailed description, upload supporting evidence where available, and submit reports either anonymously or with their identity.

The system aims to improve communication between the campus community and security personnel, encourage timely reporting of incidents, and contribute to creating a safer learning environment.

1.3 Intended Audience

This document is intended for:

- Students
 - Academic Staff
 - Non-academic Staff
 - Landlords
 - Security Personnel
 - Software Developers
 - Project Supervisor
 - System Testers
-

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
SIM	Security Incident Mapping
SRS	Software Requirements Specification
UML	Unified Modeling Language
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JavaScript	Programming language used to create interactive web pages
Admin	System Administrator

1.5 References

The development of this document is based on:

- IEEE Software Requirements Specification Guidelines
 - Google Forms Survey Conducted for the SIM Project
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2. OVERALL DESCRIPTION

2.1 Product Perspective

The Security Incident Mapping (SIM) System is a standalone web application designed to simplify the reporting of security incidents within the campus environment. It serves as a centralized platform where members of the campus community can report incidents without visiting the security office physically.

The application is developed using HTML, CSS, and JavaScript and demonstrates software engineering concepts such as requirements analysis, system modeling, user interface design, and software documentation.

2.2 Product Functions

The SIM system provides the following major functions:

- Report security incidents.
 - Select incident category.
 - Specify incident location.
 - Describe incident details.
 - Upload supporting evidence.
 - Report anonymously.
 - Submit incident reports.
 - Generate a unique incident reference number.
 - Display submission confirmation.
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2.3 User Classes

Students

Students use the application to report incidents they witness or experience.

Staff

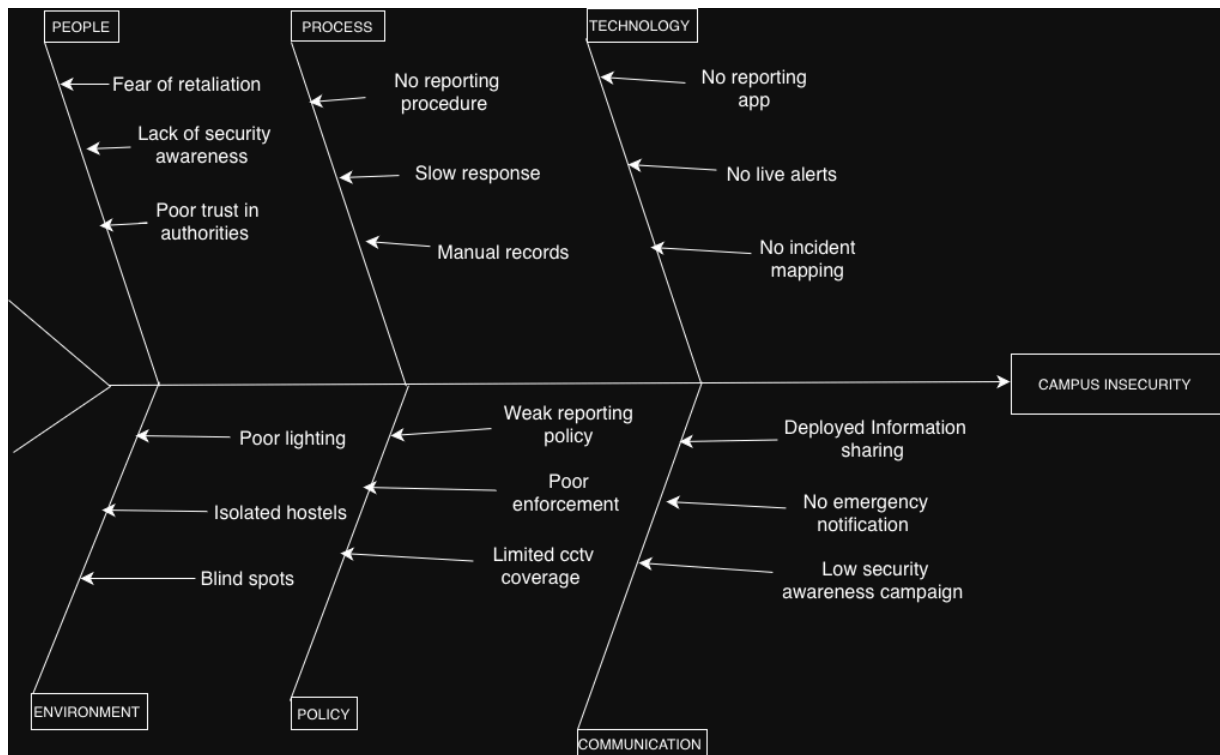
Staff members report incidents involving students, employees, or campus property.

Landlords

Landlords report incidents occurring within student hostels or residential properties around campus.

Administrator

The administrator monitors reports, reviews submitted incidents, and manages records.



To understand the major causes of campus insecurity, an Ishikawa (Fishbone) Diagram was developed. The diagram identifies the root causes under six major categories: People, Process, Technology, Environment, Policy, and Communication. The analysis guided the identification of the system requirements for the proposed Security Incident Mapping (SIM) system.

2.4 Operating Environment

Hardware

- Desktop Computer
- Laptop
- Smartphone
- Tablet

Software

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

Development Tools

- HTML5
 - CSS3
 - JavaScript
 - Visual Studio Code
 - Draw.io
 - GitHub
-

2.5 Design Constraints

The system shall:

- Operate without a database.
 - Be developed using only HTML, CSS, and JavaScript.
 - Run in a web browser.
 - Be responsive across different screen sizes.
 - Be simple enough for demonstration purposes.
-

2.6 Assumptions and Dependencies

It is assumed that users have:

- Internet access.
- A modern web browser.
- Basic knowledge of using web applications.

The system depends on:

- Browser support for JavaScript.
- Stable internet connection when hosted online.

3. SURVEY ANALYSIS

A questionnaire was created using Google Forms to understand the security challenges experienced within and around the campus community.

The survey identified **theft/stealing** as the most frequently experienced security incident among respondents. **Physical assault or stabbing** was the second most common incident, while **sexual assault** and **break-ins/burglary** were also reported. No respondents indicated experiencing cultism or gang violence in the collected responses.

These findings support the need for a centralized reporting platform that enables members of the campus community to report incidents quickly and efficiently. The results also influenced the design of the SIM system by ensuring that the most common incident types are available within the reporting form and that users have the option to submit reports anonymously.

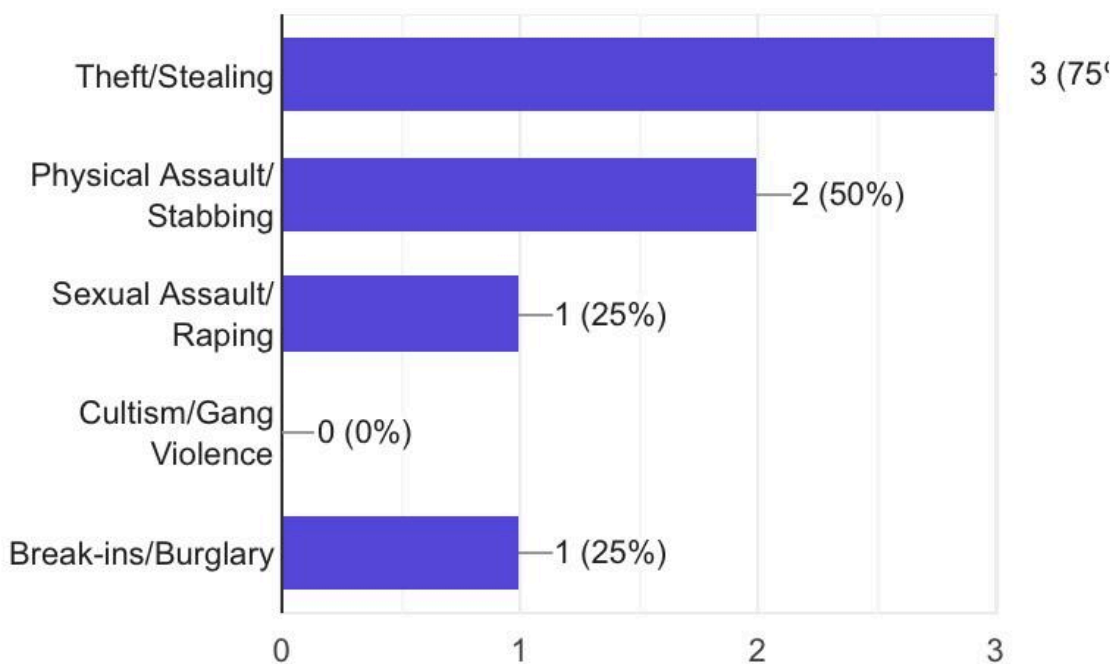
Untitled form

Questions **Responses** 4 Settings

Which of these security incidents have you witnessed or experienced on or around campus?(Select all that apply)

 Copy chart

4 responses



How do you currently report security incidents ?

 Copy chart

4 responses

4. Functional Requirements

Functional requirements describe the services and operations that the SIM system must perform. Each requirement is assigned a unique identifier for easy reference.

Requirement ID	Functional Requirement	Description
FR-001	User Access	The system shall allow students, staff, and landlords to access the application through a web browser without requiring login credentials.
FR-002	Incident Reporting	The system shall allow users to report security incidents using an online form.
FR-003	Incident Category	The system shall allow users to select the type of incident from a predefined list (Theft, Assault, Sexual Assault, Burglary, Harassment, Others).
FR-004	Incident Location	The system shall allow users to enter the location where the incident occurred.
FR-005	Incident Description	The system shall allow users to provide a detailed description of the incident.
FR-006	Anonymous Reporting	The system shall provide users with the option to submit reports anonymously.
FR-007	Evidence Upload	The system shall allow users to upload supporting evidence such as photographs or documents where available.
FR-008	Form Validation	The system shall validate mandatory fields before allowing submission.
FR-009	Report Submission	The system shall submit the incident report successfully after validation.
FR-010	Reference Number Generation	The system shall automatically generate a unique reference number for every submitted report.
FR-011	Confirmation Display	The system shall display a confirmation message after successful submission.

FR-012	Administrator Access	The system shall allow the administrator to review submitted incident reports (conceptually represented in the use case).
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5. Non-Functional Requirements

Non-functional requirements describe the quality attributes expected from the system.

5.1 Performance

The system should respond to user interactions within **3 seconds** under normal conditions.

The application should process submitted reports without noticeable delay.

5.2 Reliability

The application should consistently accept and process valid reports.

Submitted information should not be lost during normal operation.

5.3 Usability

The interface should be simple and easy to navigate.

Users with little or no technical experience should be able to complete a report without assistance.

Buttons, menus, and forms should be clearly labelled.

5.4 Security

The application should support anonymous reporting.

Sensitive user information should not be unnecessarily collected.

Input validation should reduce invalid submissions.

5.5 Maintainability

The source code should be organized into separate HTML, CSS, and JavaScript files.

The system should be easy to update with additional features in future versions.

5.6 Compatibility

The system should function correctly on:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Safari

It should also display correctly on desktops, laptops, tablets, and smartphones.

6. External Interface Requirements

6.1 User Interface

The user interface consists of:

- Navigation bar
- Home page
- About section
- Incident reporting form
- Confirmation page
- Footer

The interface uses responsive design to ensure compatibility across different devices.

6.2 Software Interface

The system is developed using:

- HTML5
- CSS3
- JavaScript

Development tools include:

- Visual Studio Code
 - Draw.io
 - GitHub
-

6.3 Hardware Interface

The system supports:

- Desktop Computers
- Laptops
- Smartphones
- Tablets



[Home](#) [About](#) [Report](#)

Security Incident Mapping

Protecting Our Campus Together.
Report security incidents quickly,
securely and anonymously.

[Report an Incident](#)

Website homepage



[Home](#) [About](#) [Report](#)

Report Security Incident

Bankole happiness

Student



Theft



Campus area

Phone was stolen

Completed report form

http://localhost:9898/Local/Srs.Sim/index.html?B5...



[Home](#) [About](#) [Report](#)



Report Submitted Successfully

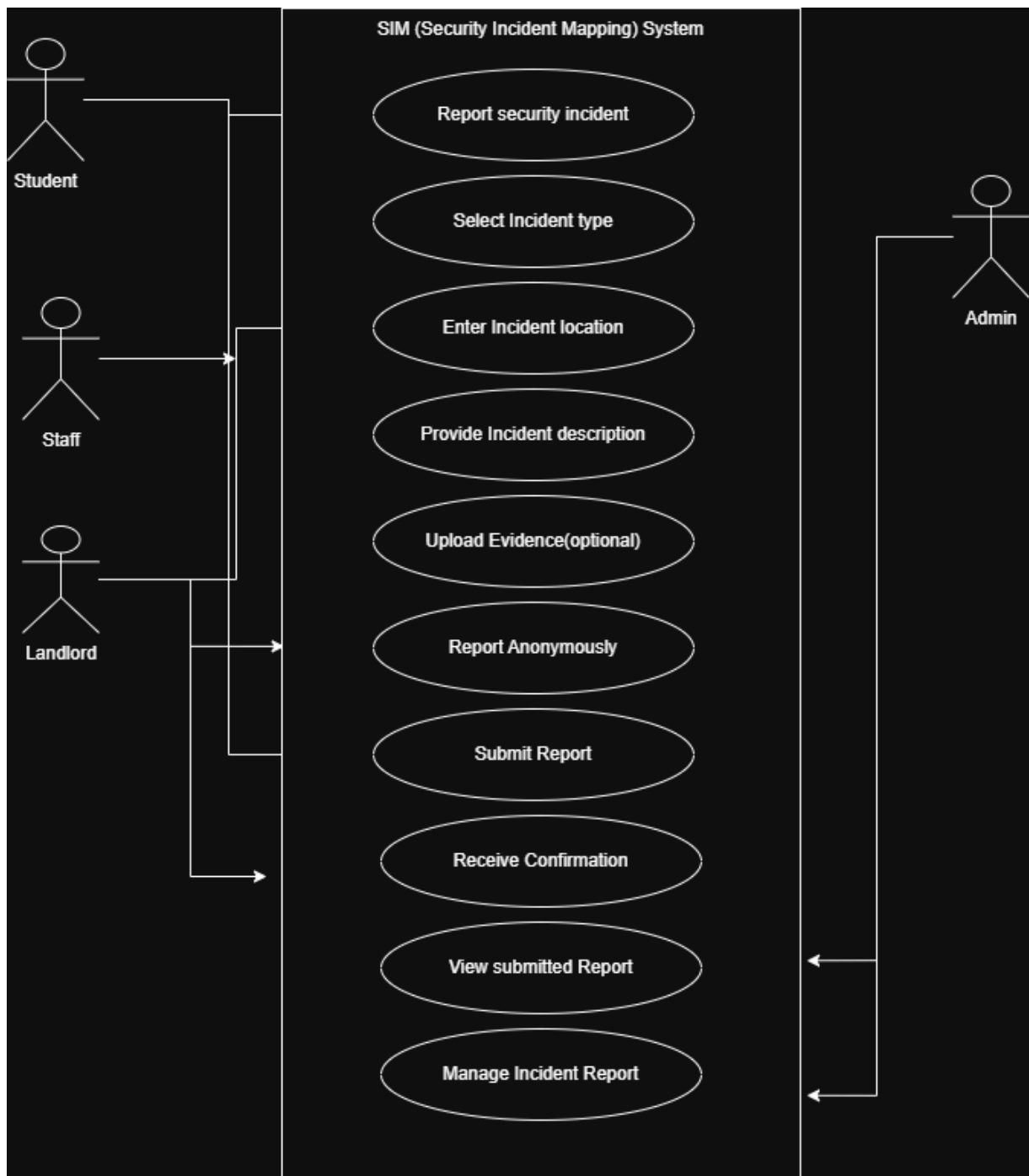
Reference Number

**SIM-2026-
9018**

Thank you for helping make
our campus safer.

[Report Another Incident](#)

7. Use Case



Descriptions

Use Case 1: Report Security Incident

Item	Description
Use Case ID	UC-001
Use Case Name	Report Security Incident
Primary Actor	Student, Staff, Landlord
Goal	Submit a security incident report
Preconditions	User has access to the SIM application
Main Flow	Open application → Fill incident form → Upload evidence (optional) → Submit report
Postcondition	Report is successfully submitted and a reference number is generated

Use Case 2: Report Anonymously

Item	Description
Use Case ID	UC-002
Primary Actor	Student, Staff, Landlord
Goal	Submit report without revealing identity
Preconditions	User selects Anonymous option
Postcondition	Incident is stored without personal identification

Use Case 3: View Submitted Reports

Item	Description
Use Case ID	UC-003
Primary Actor	Administrator

Goal	View submitted reports
Preconditions	Reports have been submitted
Postcondition	Administrator reviews available reports

Use Case 4: Manage Reports

Item	Description
Use Case ID	UC-004
Primary Actor	Administrator
Goal	Manage reported incidents
Preconditions	Administrator accesses system
Postcondition	Reports are reviewed for appropriate action

8. Object Listing

The SIM system consists of several important objects that work together to achieve the objectives of the application.

Object	Description	Attributes
User	Individual using the system	Name (optional), Role
Student	Student reporting an incident	Student Role
Staff	Staff member reporting an incident	Staff Role
Landlord	Landlord reporting an incident	Landlord Role
Incident Report	Stores incident information	Incident Type, Description, Date, Time
Location	Specifies where the incident occurred	Building, Hostel, Street, Area

Evidence	Supporting file uploaded by user	Image/File
Anonymous Report	Indicates anonymous submission	Anonymous Status
Reference Number	Unique report identifier	Report ID
Administrator	Reviews reports	Admin ID, Reports Managed

9. Assumptions and Constraints

Assumptions

The following assumptions were made during system development:

- Users have internet access.
 - Users possess basic computer or smartphone skills.
 - Users provide accurate incident information.
 - Modern web browsers support JavaScript.
 - The system will primarily be used within the campus environment.
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Constraints

The project is subject to the following constraints:

- Developed using only HTML, CSS, and JavaScript.
 - No database integration.
 - No user authentication module.
 - Prototype developed for academic purposes.
 - Administrator functions are represented conceptually through the use case diagram and are not fully implemented.
-

10. Future Enhancements

Future versions of the SIM system may include:

- User login and authentication.
- GPS-based incident location mapping.
- Integration with campus security databases.
- Email and SMS notifications.
- Administrator dashboard.
- Incident status tracking
- Mobile application version.

11. Conclusion

The **Security Incident Mapping (SIM)** system was proposed to address the growing concern of security incidents within and around the campus community. The survey conducted as part of this project revealed that **theft/stealing** was the most commonly experienced security incident among respondents, followed by **physical assault**, while **sexual assault** and **break-ins/burglary** were also reported. These findings indicate the need for a faster, more accessible, and centralized method of reporting security incidents.

The analysis carried out using the **Ishikawa (Fishbone) Diagram** identified several root causes contributing to campus insecurity, including inadequate reporting procedures, poor communication, weak security policies, limited technological support, environmental factors, and low security awareness. These factors highlighted the need for a solution that would simplify and encourage incident reporting.

To address these challenges, the Security Incident Mapping (SIM) system was designed as a web-based application that enables students, staff, and landlords to report security incidents quickly and conveniently. The system provides features such as incident categorization, location specification, detailed incident descriptions, optional evidence uploads, anonymous reporting, and automatic generation of a unique reference number upon successful submission.

The software requirements presented in this document define the functional and non-functional requirements necessary for the successful development of the system. The use case diagram, object listing, and system requirements provide a clear representation of how users interact with the application and how the system is expected to operate.

Although the current implementation serves as a prototype developed for academic purposes, it demonstrates the practical application of software engineering principles, including requirements analysis, system modelling, user interface design, and software documentation. With future enhancements such as user authentication, database integration, GPS-based incident mapping, real-time notifications, and an administrative dashboard, the SIM system has the potential to become a comprehensive campus security management solution.

In conclusion, the Security Incident Mapping (SIM) system provides a practical and scalable approach to improving campus safety by encouraging timely reporting, enhancing communication between users and security personnel, and supporting informed decision-making for campus security management.